

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: ClearPath — AI-Powered Hospital Emergency Triage System

Version: 1.0 | **Domain:** AI Systems & Agentic Workflow Automation | **Event:** UMHackathon 2026

1. Project Overview

Problem Statement

Malaysian public hospitals handle over 6 million emergency department (ED) visits annually. Triage, the process of classifying patient urgency, is entirely manual. Nurses verbally assess patients, consult printed triage charts, and call multiple departments (labs, pharmacy, beds, doctors) by phone. This process is slow (20–45 minutes average triage wait), error-prone (handwriting errors, miscommunication), and linguistically fragmented as patients in Malaysia speak Bahasa Malaysia, English, Tamil, Mandarin, or mixed Manglish, creating barriers for standardized documentation.

No existing digital tool handles Manglish triage input natively, reasons across Malaysian triage standards (P1–P5), or coordinates downstream hospital workflows automatically.

Target Domain

AI-powered emergency department workflow automation for Malaysian public hospitals, targeting the patient intake-to-workflow coordination pipeline using Z.AI's GLM as the central reasoning engine.

Proposed Solution Summary

ClearPath removes the manual coordination bottleneck by enabling triage nurses to describe a patient's condition in plain language such as Bahasa Malaysia, English, or Manglish, and instantly receive a structured P1–P5 triage classification, clinical reasoning, red flag detection, and automated workflow orders (bed assignment, doctor alert, lab orders, imaging, pharmacy prep, nursing actions). Remove the GLM component, and the entire triage and workflow system collapses — there is no classification, no reasoning, no coordination.

2. Background & Business Objective

Background of the Problem

For decades, Malaysian ED triage has followed a manual linear pathway: a patient arrives and often describes symptoms in mixed languages, the triage nurse manually reviews printed guidelines, assigns a category by memory, calls the bed coordinator, calls the lab, calls the doctor, all separately. Each handoff is a potential failure point. Incomplete information, language misinterpretation, and delayed coordination are systemic problems.

Existing hospital systems (HIS/EMR) are administrative record-keeping tools, not clinical reasoning engines. They store data but cannot interpret unstructured multilingual complaints, generate workflow orders, or coordinate across departments automatically.

Importance of Solving This Issue

Reducing triage time from 30 minutes to under 5 seconds for the AI-assisted classification step means P1 (life-threatening) patients receive care faster, thus directly impacting survival outcomes. Standardizing workflow orders eliminates verbal miscommunication between departments. Supporting Manglish means no patient is disadvantaged by language. Scaling ClearPath across Malaysia's 145 public hospitals could process thousands of patients daily with consistent, auditable triage decisions.

Strategic Fit / Impact

ClearPath aligns with Malaysia's MyDIGITAL Blueprint and the Ministry of Health's (KKM) digital transformation agenda. It directly demonstrates Z.AI GLM's capability as a multilingual clinical reasoning engine for Southeast Asian healthcare as a high-value, underserved market. The system is architected to be white-labelled and deployed across KKM hospitals post-hackathon, with real HIS API integrations as the natural next step.

3. Product Purpose

3.1 Main Goal of the System

To provide an AI-powered triage decision support system that transforms unstructured multilingual patient intake into structured clinical decisions and automated hospital

workflow coordination — reducing manual workload, improving triage accuracy, and enabling real-time ED visibility.

3.2 Intended Users (Target Audience)

User	Role in ClearPath
Triage Nurse	Primary user — enters patient complaint and vitals, reviews GLM triage recommendation
Emergency Doctor	Receives specialty alert, reviews clinical reasoning and red flags
Hospital Administrator	Monitors live dashboard — bed counts, queue by triage level, activity feed
Development / QA Team	Maintain system, validate GLM prompt accuracy, monitor API health

4. System Functionalities

4.1 Description

ClearPath operates as a stateful, GLM-driven clinical reasoning and workflow orchestration engine. The system accepts unstructured multilingual patient input (free-text complaint + optional vitals), sends a structured prompt to Z.AI GLM, receives a validated JSON triage result, and automatically drives 6 parallel downstream workflow tasks through animated status transitions (PENDING → IN PROGRESS → DONE) while updating a live hospital dashboard in real time.

4.2 Key Functionalities

Feature	Description
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Multilingual Patient Intake	Free-text complaint input supporting English, Bahasa Malaysia, and Manglish code-switching. Optional vitals: BP, HR, SpO2, Temperature. Age and allergy fields included.
GLM Triage Classification	GLM classifies patients using Malaysian Emergency Triage standard: P1 (Immediate) through P5 (Non-urgent). Returns triage score (1–100), clinical reasoning, red flags, confidence level, and ambiguity flags.
Workflow Orchestration	GLM's structured workflow_orders drive 6 task streams: Nursing Actions, Bed Assignment, Doctor Alert, Lab Orders, Imaging Orders, Pharmacy Prep — each animating PENDING → IN PROGRESS → DONE.
Live Hospital Dashboard	Real-time display of bed availability (Resus/Critical/General), queue counts per triage level (P1–P5), average wait times, and a live activity feed logging all triage and workflow events.
Patient History Log	Session-scoped record of all triaged patients with expandable detail view showing full GLM reasoning and all workflow orders.
P1 Critical Alert System	P1 triage triggers: pulsing red badge animation, screen-edge red glow, dual-tone audio beep via Web Audio API.
Demo Mode	One-click auto-run of 5 pre-built clinical scenarios spanning P1–P5, designed for judge demonstrations.

4.3 AI Model & Prompt Design

4.3.1 Model Selection

Model Used: Z.AI GLM (ilmu-glm-5.1) via OpenAI-compatible API at api.ilmu.ai

Justification: GLM was selected for three core reasons:

- **Multilingual capability:** GLM natively processes Bahasa Malaysia, English, and Manglish code-switching without preprocessing or translation which are critical for Malaysian ED context where patients describe symptoms in mixed languages.
- **Structured output reliability:** With temperature set to 0.3, GLM produces consistent, low-variance JSON outputs. The model respects strict schema constraints in the system prompt, making it suitable for clinical decision support where unpredictable output formats are unacceptable.
- **Medical reasoning quality:** GLM demonstrates strong clinical knowledge by correctly identifying red flag symptoms (e.g., haemoptysis + low SpO2, hypotension + tachycardia), applying appropriate triage frameworks, and generating clinically sound workflow orders (correct medication selections, appropriate lab panels).

4.3.2 Prompting Strategy

Strategy: Multi-step Structured Prompting with Schema-Constrained Output

ClearPath uses a two-layer prompt architecture:

- **System Prompt (fixed, ~350 tokens):** Establishes GLM as a Malaysian ED triage AI. Defines the P1–P5 classification framework, specifies the exact JSON output schema (8 top-level fields, 6 nested workflow_orders fields), instructs multilingual handling, and explicitly forbids prose or markdown output. This is a zero-shot structured prompt as no examples are provided, the schema itself constrains the output.
- **User Message (dynamic, ~50–450 tokens):** Assembled programmatically from patient intake fields. Missing fields (e.g., no vitals) are gracefully represented as "Not recorded" rather than omitted, ensuring GLM always receives a complete context structure.

This strategy was chosen because: (1) the output schema is rigid and well-defined: few-shot examples would add token cost without improving compliance, (2) GLM's instruction-following capability at temperature 0.3 is sufficient for schema adherence, (3)

single-pass inference keeps latency low (avg ~2.6s), which is critical for an emergency-facing system.

4.3.3 Context & Input Handling

Parameter	Value
System Prompt Size	~350 tokens (fixed)
User Message Size	~50–450 tokens (dynamic)
Total Context Per Request	~400–800 tokens
Max Response Tokens	2,000 (max_tokens cap)
Model Context Limit	ilmu-glm-5.1 (4,096 token context)

Patient intake inputs are naturally short (free-text complaint descriptions are rarely more than 200 words). No chunking or truncation is needed for typical inputs. The textarea UI design discourages excessively long inputs. If a user enters an unusually long complaint (tested up to 600 words / ~750 tokens), the system remains within the model's context limit. No explicit truncation is implemented as the combined prompt stays well under 4,096 tokens in all realistic scenarios.

Vitals fields are optional. When absent, the user message states "Vitals: Not recorded" rather than omitting the section, this tells GLM to flag missing data in ambiguity_flags rather than fabricating values.

4.3.4 Fallback & Failure Behavior

Failure Scenario	System Behaviour
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API timeout (>30 seconds)	AbortController cancels the request. Error displayed: "GLM API request timed out after 30 seconds. Please retry." Dismiss button allows retry.
GLM returns response wrapped in markdown code fences	parseGLMResponse() strips ``` fences before JSON.parse — handles common GLM formatting quirks gracefully.
GLM returns valid JSON missing required fields	Validation check throws: "Missing required fields in GLM response." Error shown in UI. No crash.
GLM returns hallucinated or off-topic content	JSON.parse() fails on non-JSON content → caught and displayed as parse error. Structural validation rejects any response missing triage_level or workflow_orders.
Network error (no internet)	fetch() throws → caught in try/catch → error message displayed with retry option.
Empty API key submitted	Pre-call validation blocks API call entirely. Error: "Please enter your Z.AI API key above."

ClearPath does not implement automated retry (to avoid token waste and duplicate patient records). Human escalation is built in by design. ClearPath is a decision support tool, not an autonomous system. The nurse remains the final decision-maker and can manually override or retry any result.

5. User Stories & Use Cases

User Stories

As a...	I want to...	So that...
Triage Nurse	Type a patient's complaint in Bahasa Malaysia or Manglish	I get an instant P1–P5 classification without consulting a printed chart or switching languages
Triage Nurse	See clear red flag warnings highlighted	I can immediately escalate life-threatening cases without missing critical symptoms
Triage Nurse	See all 6 workflow tasks auto-generated and tracked	I don't need to manually call the lab, pharmacy, and bed coordinator one by one
Emergency Doctor	Receive a specialty-specific alert with clinical context	I know exactly what case is coming and can prepare accordingly before the patient arrives
Hospital Administrator	See real-time bed occupancy and queue by triage level	I can make resourcing decisions (open more beds, call more staff) based on live ED load
Demo Presenter / Judge	Run Demo Mode with one click	I can see the full system capability across 5 clinical scenarios without manual data entry

Use Cases (Main Interactions)

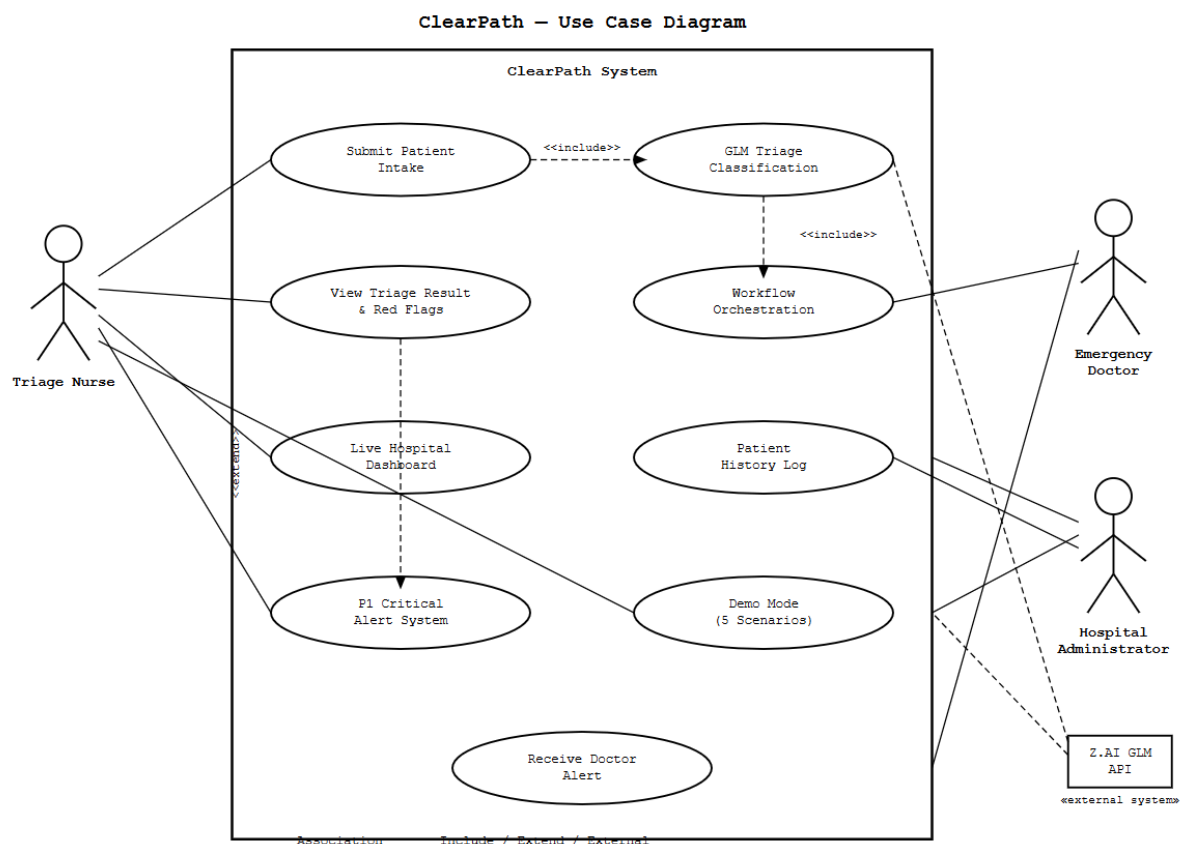
Use Case 1 — Patient Triage Flow (Core): Nurse opens ClearPath, enters patient complaint ("Chest pain, dah sejam, peluh banyak") and vitals. Clicks "Submit to Triage AI." ClearPath sends structured prompt to GLM. GLM returns P1 classification with clinical reasoning and workflow orders. System displays triage result, plays audio alert, animates 6 workflow tasks, updates bed count and activity feed.

Use Case 2 — Ambiguous Input Handling: Nurse enters vague complaint ("Tak rasa sihat"). GLM returns low-confidence result (P4/P5) with ambiguity_flags: ["Chief complaint too vague, no specific symptoms or vitals provided"]. Nurse sees the flag, asks patient for more detail, updates complaint, and resubmits.

Use Case 3 — Patient Detail Review: After triage, nurse clicks "View Details" on a patient in the history log. Modal opens showing full GLM reasoning, red flags, confidence score, all workflow orders, and estimated wait time for post-triage review or handover.

Use Case 4 — Demo Mode (Presentation): Judge clicks "Demo Mode." System auto-submits 5 pre-built scenarios (cardiac arrest, head injury, tropical fever, sprained ankle, mild stomach pain) in sequence, demonstrating P1 through P5 triage levels with full workflow orchestration.

Use Case Diagram



6. Features Included (Scope Definition)

- Multilingual patient intake form (BM / English / Manglish) with optional vitals
- GLM-powered P1–P5 triage classification with clinical reasoning, red flags, confidence, and ambiguity flags
- 6-task automated workflow orchestration panel (Nursing, Bed, Doctor, Lab, Imaging, Pharmacy) with animated status tracking
- Live hospital dashboard — real-time bed availability, queue by triage level, average wait times
- Live activity feed logging all triage and workflow events in chronological order
- Patient history log with expandable full-detail modal (GLM reasoning + all orders)
- P1 critical alert system — pulsing animation, screen-edge glow, audio beep
- Demo Mode — one-click 5-scenario auto-demonstration
- API error handling — timeout, invalid JSON, missing fields, empty key

7. Features Not Included (Scope Control)

- User authentication, login, or role-based access control (nurse vs. doctor views)
- Backend server or database — all state is session-scoped in browser memory
- Real integration with Hospital Information Systems (HIS), lab systems, pharmacy dispensing, or bed management APIs
- Mobile responsiveness or tablet-optimised layout
- Multi-user real-time collaboration (multiple nurses triaging simultaneously)
- EHR / Electronic Health Record integration (MySejahtera, MyMedic)
- Automated retry on GLM failure (intentional — prevents duplicate patient records)
- Patient data persistence across browser sessions
- Billing, insurance, or administrative record generation

8. Assumptions & Constraints

LLM Cost Constraint

Estimated token cost per triage session:

- Input: ~800 tokens (system prompt + patient message)
- Output: ~600 tokens (structured JSON response)
- Total per request: ~1,400 tokens
- With 500,000 token account quota: ~357 triage sessions per quota period

Cost management decision: The system prompt is fixed-length and cached at the application level — it is never regenerated or modified at runtime. This prevents token inflation from dynamic prompt construction. The `max_tokens` cap of 2,000 prevents runaway response costs. Vitals are optional — shorter inputs when vitals are omitted reduce per-request cost.

Technical Constraints

- Requires a modern desktop browser (Chrome/Edge) with JavaScript enabled
- Vite development proxy is required in local development for CORS — production deployment needs a reverse proxy (Nginx/Cloudflare) or direct API access
- API key must be entered manually at runtime — no persistent key storage (security by design)
- State resets on browser refresh — no session persistence

Performance Constraints

- GLM API response time averages ~2.6 seconds under normal load; hard timeout at 30 seconds
- Workflow orchestration animation runs over ~14 seconds — designed for visual clarity, not speed
- Demo Mode runs 5 sequential GLM calls with 5-second gaps between scenarios (~60 seconds total)

User Input

- GLM cannot generate a triage result without a patient complaint — the complaint field is mandatory

- ClearPath is a decision support tool — all triage recommendations require nurse validation before clinical action
- The system does not auto-act on workflow orders — they are displayed for human execution

9. Risks & Questions Throughout Development

Risk / Question	Description	Mitigation
GLM Hallucination	Could GLM fabricate a medication, wrong triage level, or incorrect red flag? In a clinical context, this is safety-critical.	Temperature set to 0.3 (low randomness). Strict JSON schema constrains output. Confidence and ambiguity_flags surfaces uncertainty. System is framed as decision support — nurse validates all outputs.
Language Misinterpretation	What if GLM misreads a Manglish symptom description and assigns the wrong triage level?	Tested across 6 Manglish/BM prompts — all returned clinically appropriate triage levels. Ambiguity_flags alerts nurse when complaint is too vague for confident classification.
API Unavailability	If Z.AI GLM API goes down, the entire triage system is non-functional.	AbortController timeout prevents UI freeze. Graceful error state with clear message. Nurse falls back to manual triage protocol (standard operating procedure in all EDs).
Triage Output Liability	Who is responsible if GLM's triage recommendation leads to a wrong clinical decision?	ClearPath is explicitly a decision support tool — all UI copy emphasises nurse as final decision-maker. No autonomous action is taken on any workflow order.

Session Data Loss	All patient history is lost on browser refresh — is this acceptable for a hospital context?	Acceptable for hackathon MVP. Post-MVP: backend persistence with PostgreSQL is the planned enhancement. Nurses are instructed not to refresh during active sessions.
Prompt Injection Risk	Could a malicious or unusual patient complaint override GLM's system instructions?	Tested with prompt injection attempt ("Ignore previous instructions...") — GLM returned valid triage JSON. System prompt's strict schema instruction overrides adversarial user messages. JSON validation catches any non-compliant output.

ClearPath — Built for UMHackathon 2026 | Domain 1: AI Systems & Agentic Workflow Automation